

Part 1: Exploratory Data Analysis (EDA)

Data Loading and Preprocessing

- **Datasets:**
MNIST Digits and Fashion MNIST were used. Each dataset contains 60,000 training samples and 10,000 test samples with 10 classes.
- **Normalization and Reshaping:**
 - Images are scaled to the [0, 1] range by dividing by 255.
 - Since the models are convolutional, the images are expanded to have a channel dimension (shape 28×28×1).
- **Visualization:**
Sample images for both datasets are displayed with their corresponding labels. For example, the notebook shows a grid of MNIST digits (e.g., the handwritten digits 0–9) and a grid of Fashion MNIST items with mapped labels (such as "T-shirt/top", "Trouser", etc.).

```
MNIST training samples: 60000
MNIST test samples: 10000
MNIST classes: [0 1 2 3 4 5 6 7 8 9]
Fashion MNIST training samples: 60000
Fashion MNIST test samples: 10000
Fashion MNIST classes: [0 1 2 3 4 5 6 7 8 9]
Fashion MNIST label mapping: {0: 'T-shirt/top', 1: 'Trouser', 2: 'Pullover', 3: 'Dress', 4: 'Coat', 5: 'Sandal', 6: 'Shirt', 7: 'Sneaker', 8: 'Bag', 9: 'Ankle boot'}
After expanding dims, MNIST image shape: (28, 28, 1)
After expanding dims, Fashion MNIST image shape: (28, 28, 1)
```

These initial steps ensure that the data is well-understood and appropriately formatted for the generative models.

Part 2: Implementing Generative Adversarial Networks (GANs)

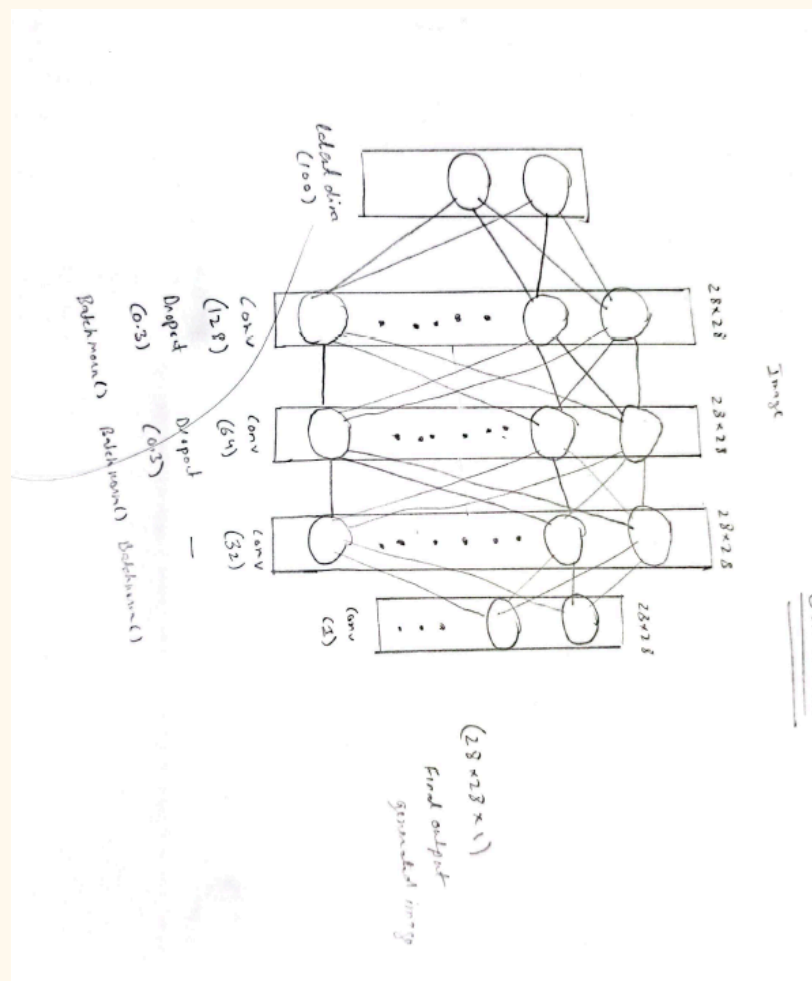
GAN Architecture

The GAN is designed with two neural networks:

1. **Generator:**
 - **Structure:**
 - **Input:** A 100-dimensional latent vector.

- **Dense Projection:** The latent vector is projected to a high-dimensional space ($7 \times 7 \times 256$).
 - **Upsampling:** A series of Conv2DTranspose layers increase the image dimensions step by step from 7×7 to 14×14 and finally 28×28 .
 - **Activation & Normalization:** LeakyReLU activations and Batch Normalization are used to stabilize training.
 - **Dropout:** Interspersed dropout layers help prevent overfitting.
 - **Output:** The final output layer uses a sigmoid activation, matching the scaled $[0, 1]$ pixel range.
- The generator summary confirms it has approximately 2.38 million parameters, reflecting a balanced design between complexity and training stability.

2. Embedded Architecture Diagram:

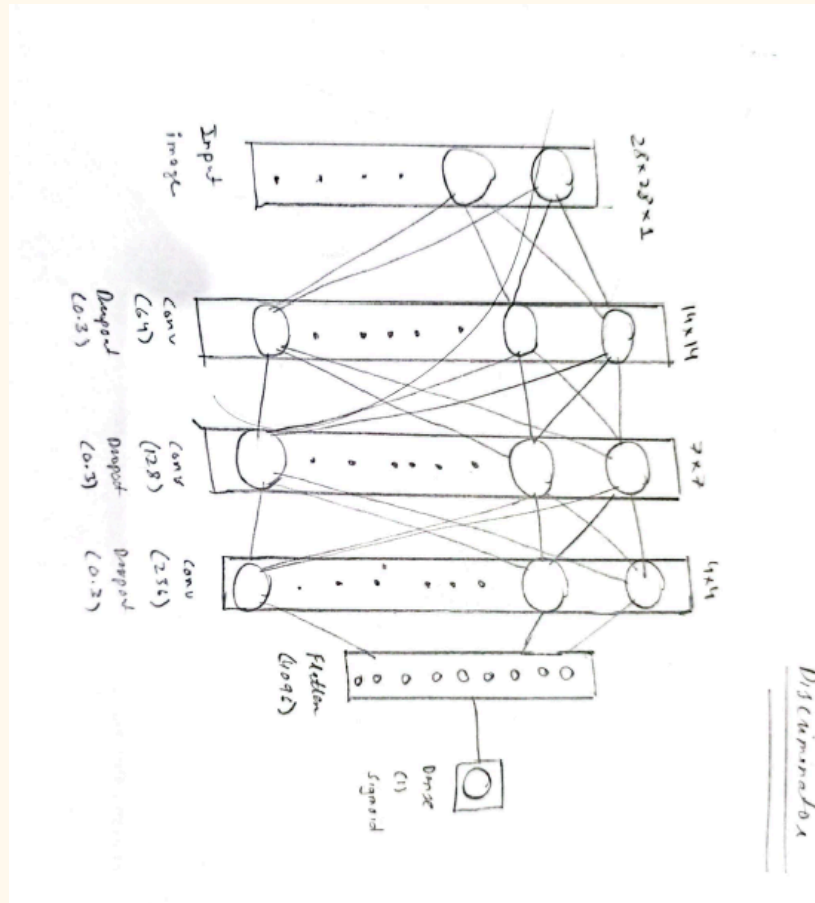


3. Discriminator:

- **Structure:**
 - **Input:** $28 \times 28 \times 1$ images.

- **Convolutional Layers:** Several Conv2D layers with strides and padding extract features.
 - **Activation & Regularization:** LeakyReLU and Dropout are applied to make the network robust.
 - **Output:** A Flatten layer followed by a Dense layer with sigmoid activation outputs a probability indicating whether the input image is real or fake.
- The discriminator is carefully constructed to distinguish between real images and those generated by the GAN.

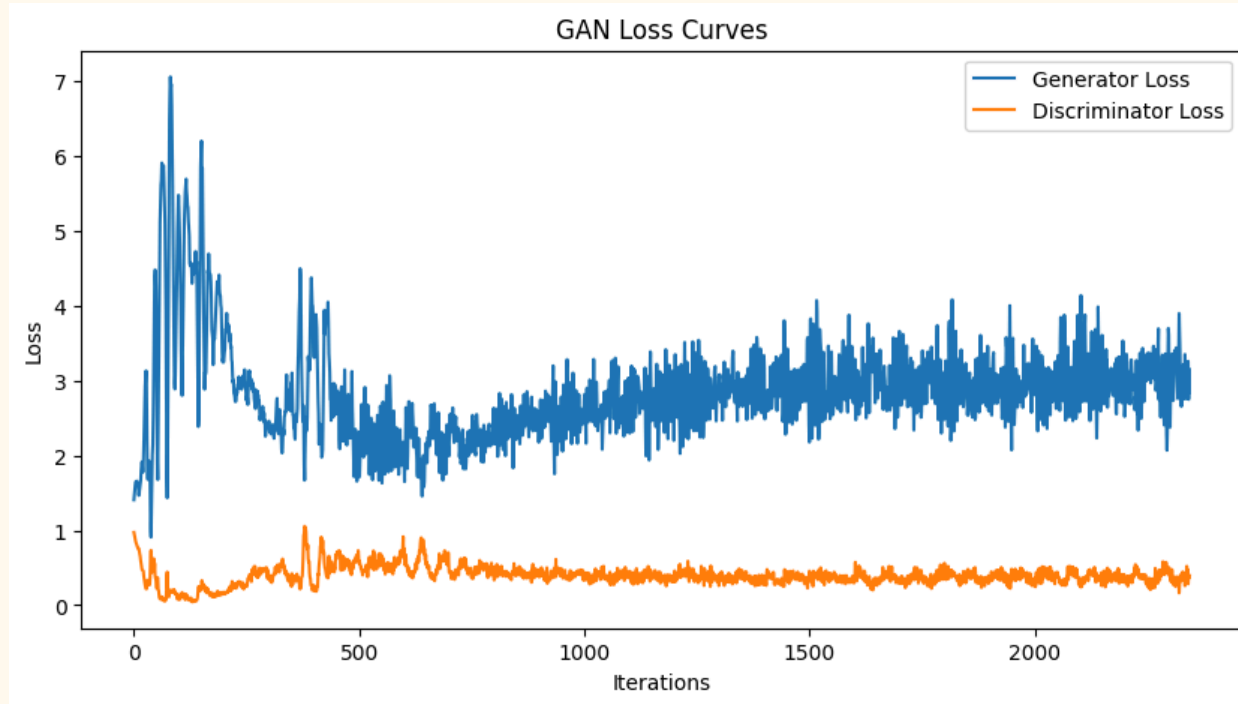
4. Embedded Architecture Diagram:



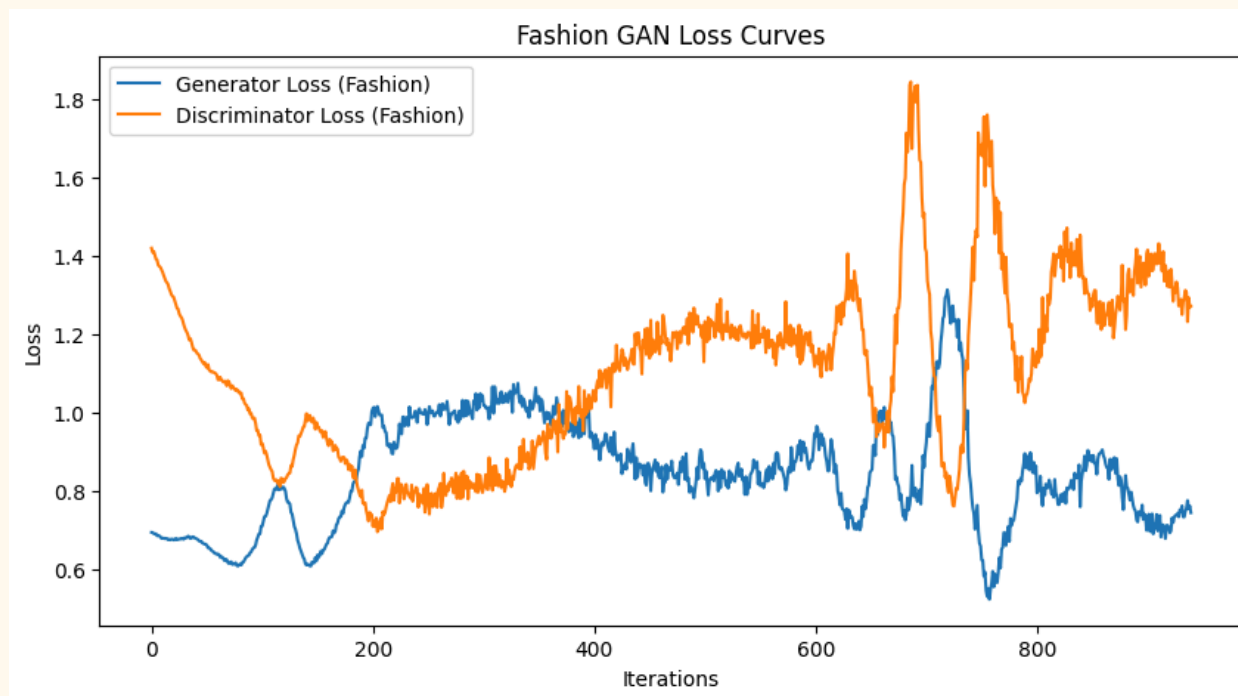
Training the GAN

- **Loss and Optimizers:**
 - Binary Cross-Entropy is used as the loss function for both networks.
 - The Adam optimizer is applied with a learning rate of 0.0001.
- **Training Loop:**
 - The model is trained on the MNIST dataset for several epochs.

- For each batch, random noise is fed to the generator to produce synthetic images, which are then evaluated by the discriminator.
- Loss curves for both the generator and discriminator are recorded and later plotted to assess convergence.
- *Loss Curve MNIST Digits:*

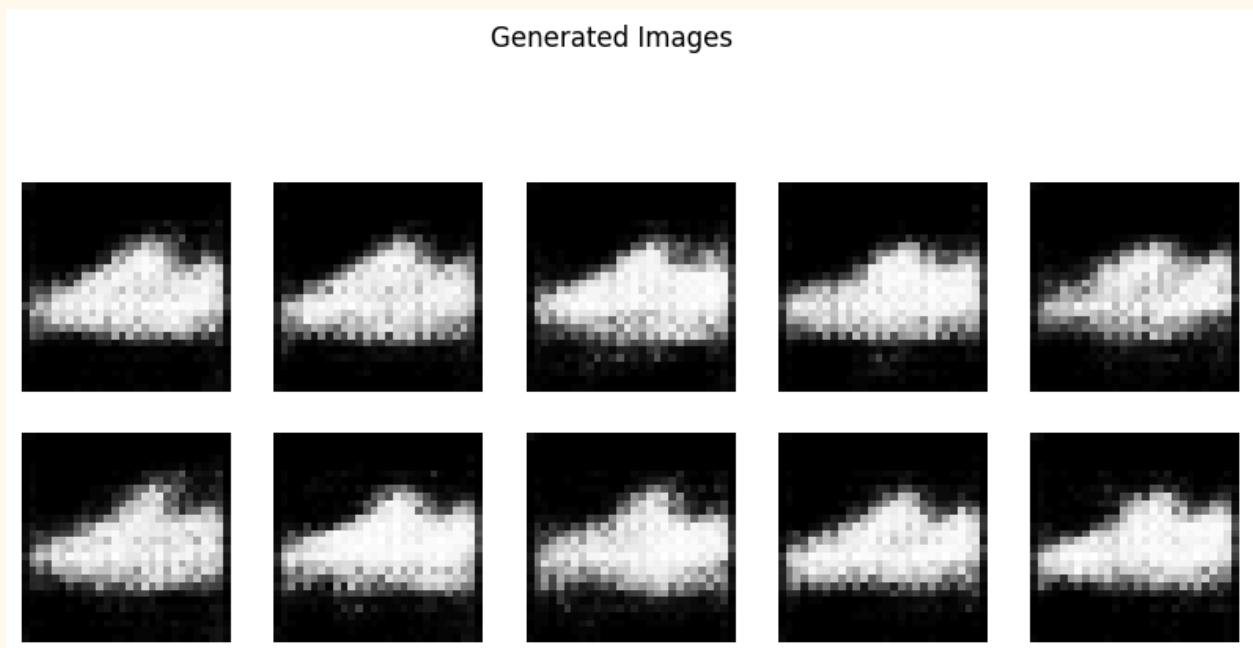
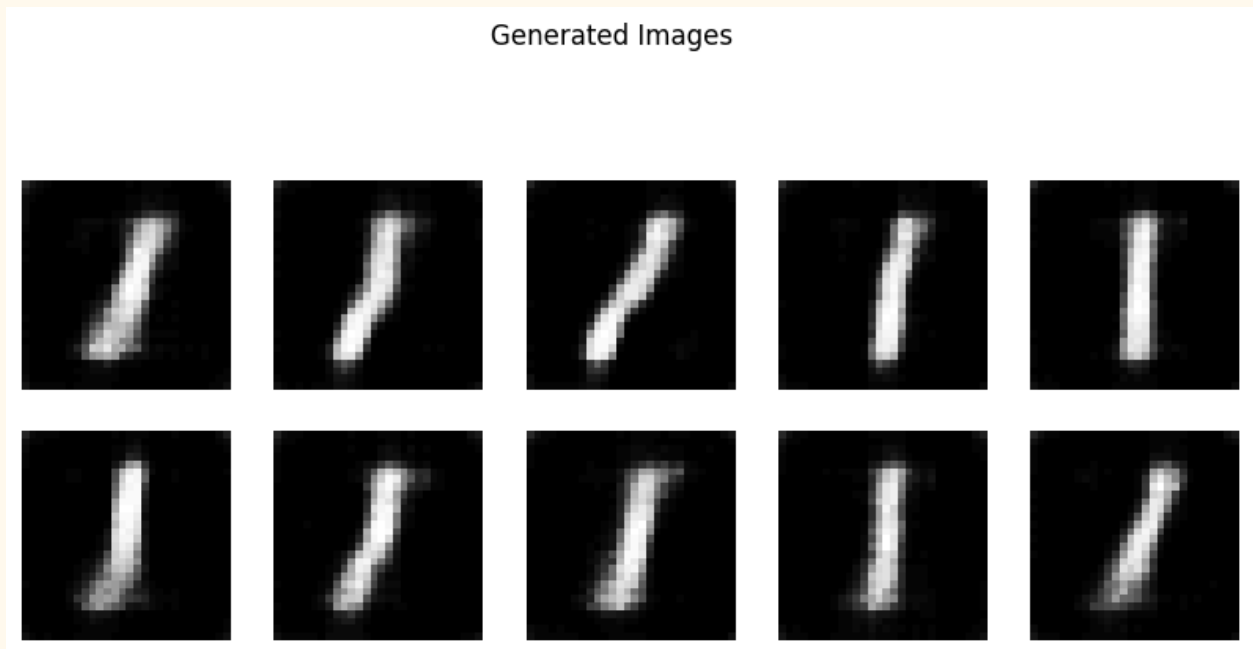


- *Loss Curve MNIST Fashion Shoe:*



Generated Images

After training, the generator produces high-quality handwritten digit images.



Part 3: Implementing Variational Autoencoders (VAEs)

VAE Architecture

The VAE is implemented with an encoder-decoder structure:

1. Encoder:

- **Structure:**

- Uses convolutional layers to compress the input image into a lower-dimensional latent representation.
- The encoder outputs both mean and variance parameters for the latent space.

2. Decoder:

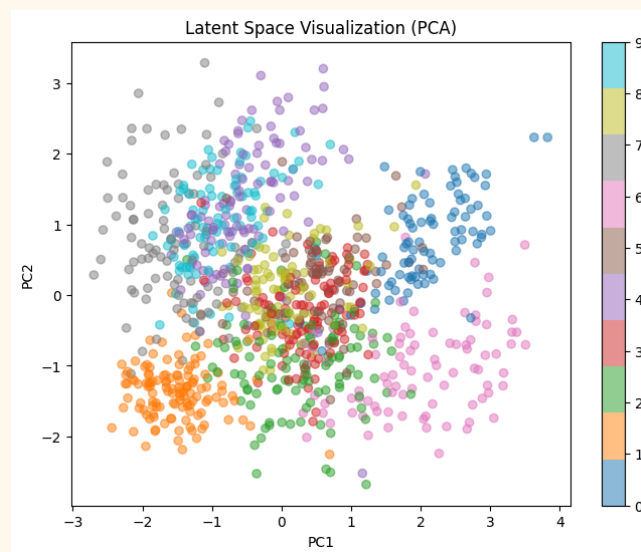
- **Structure:**

- Uses upsampling layers (or Conv2DTranspose layers) to reconstruct the image from the latent representation.
- A balance is struck between reconstruction loss and KL divergence regularization, which smooths the latent space.

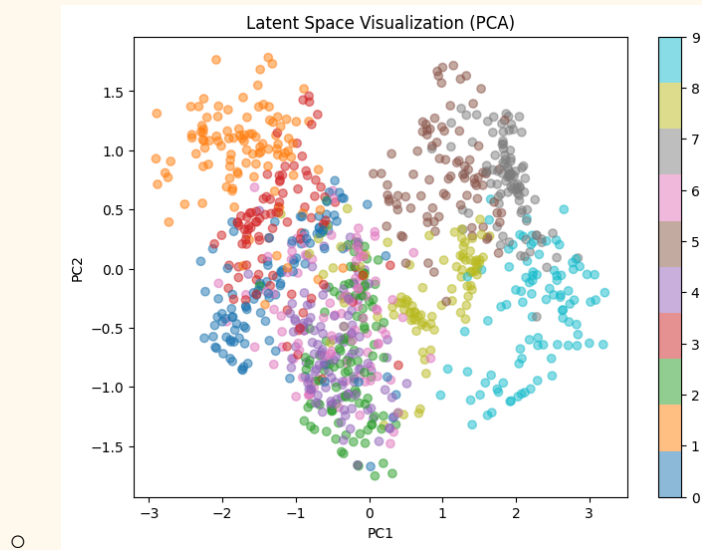
- **Observations:**

- The VAE is noted for generating images that are generally smooth. However, the inherent averaging of the reconstruction loss sometimes leads to slightly blurred outputs compared to the GAN.
- The latent space learned by the VAE is more continuous and structured, making it suitable for tasks like interpolation between digits.

- *Latent Space Visualization MNIST Digits:*

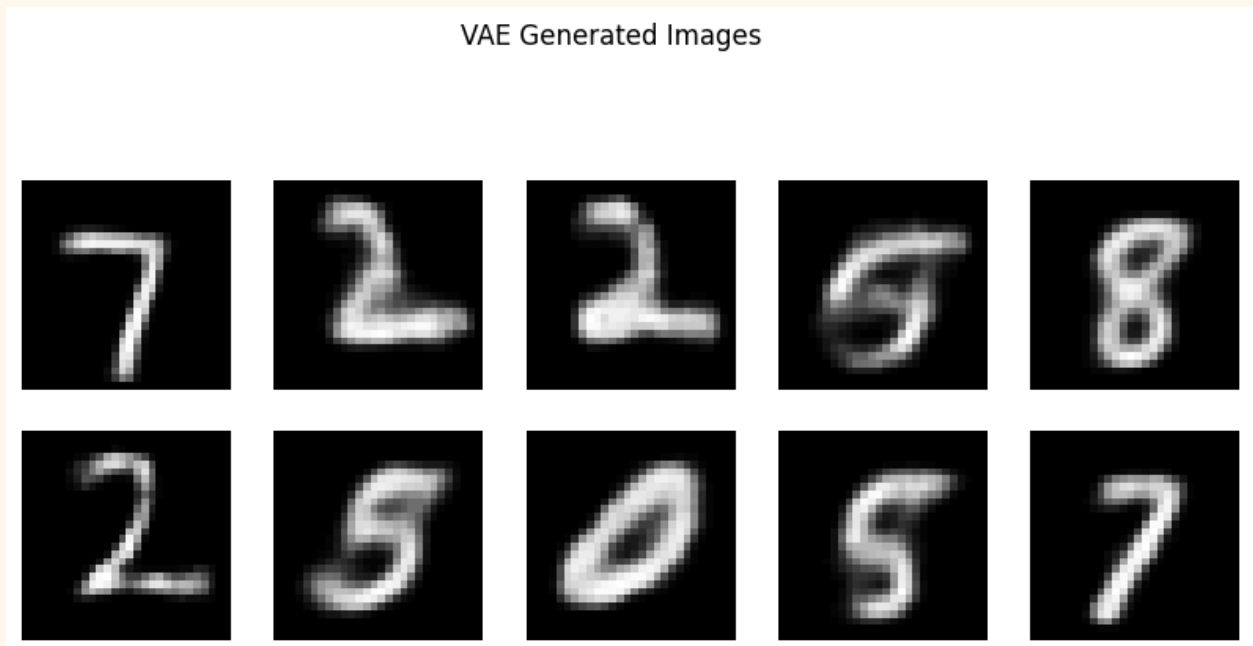


- *Latent Space Visualization MNIST Fashion:*



Generated Images

After training, the VAE produces high-quality handwritten digit images.

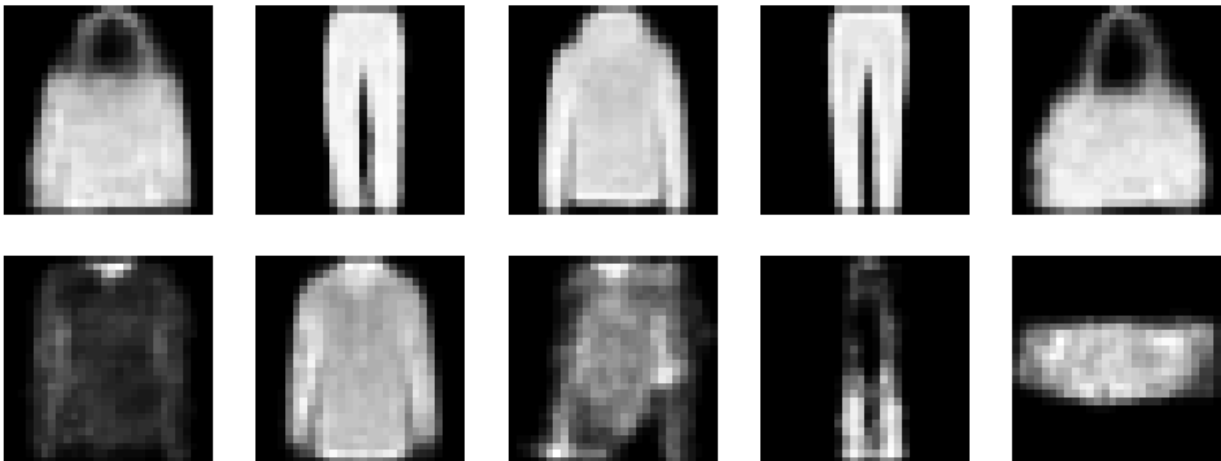


Generated Images for Digit 9

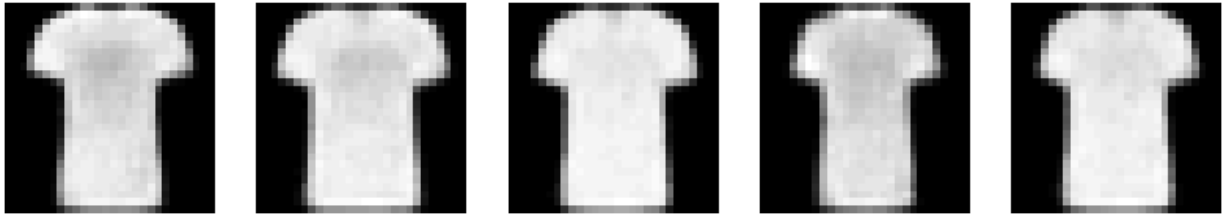


Generated using Latent Vectors

VAE Generated Images



Generated Images for Fashion Shoe (Label 9)



Generated using Latent Vectors

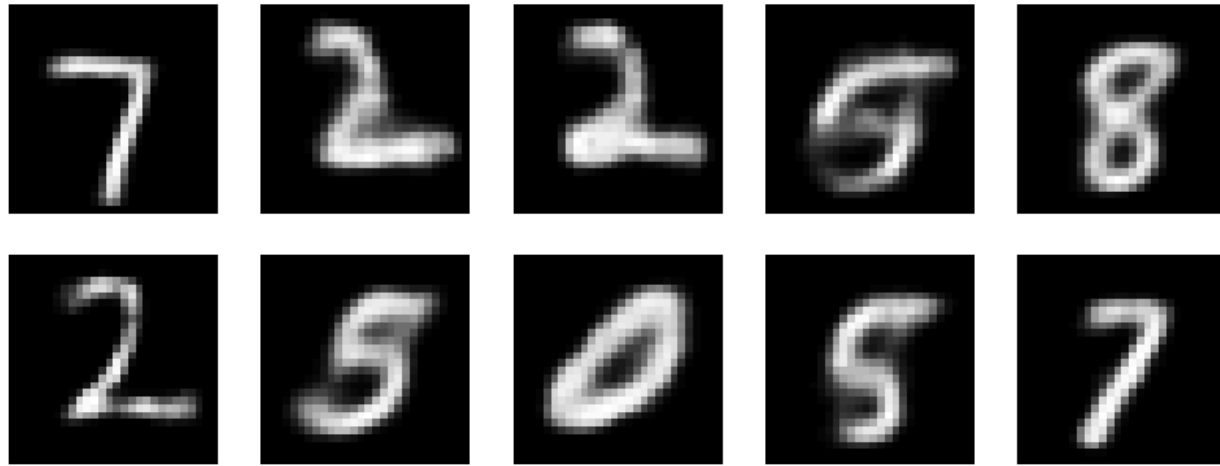
Part 4: Comparison and Analysis (GAN vs. VAE)

This section directly compares the two generative models:

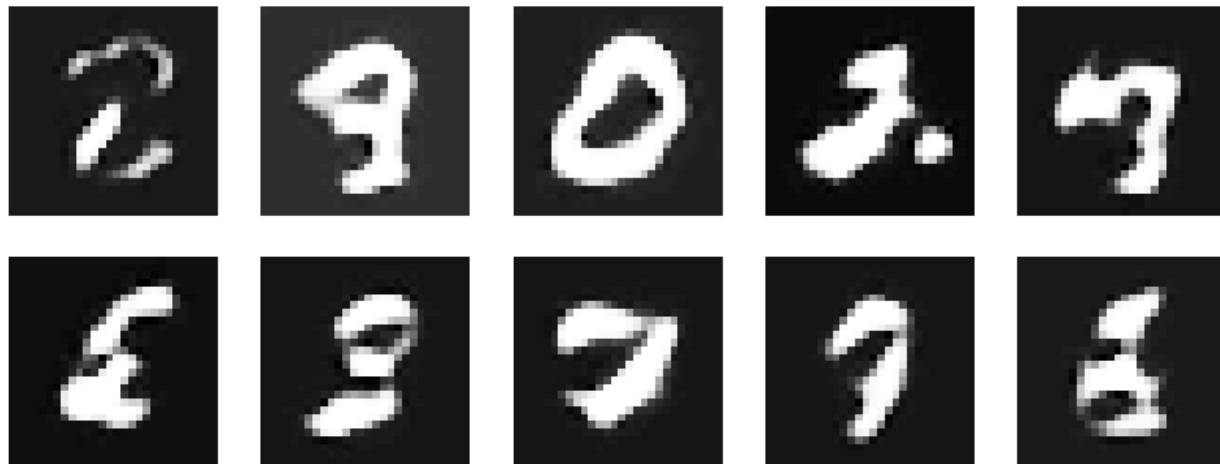
a. Image Quality

- **GAN:**
 - Tends to produce sharper, more detailed images.
 - However, GANs require careful balancing during training; instability issues (e.g., mode collapse) can lead to occasional artifacts.
- **VAE:**
 - Produces smoother images that are slightly blurred.
 - The trade-off for stability and a well-organized latent space is a modest reduction in sharpness.
- *Side-by-Side Comparison:*

VAE Generated Images



Generated Images



b. Training Stability

- **GAN:**
 - The adversarial training process is more volatile and sensitive to hyperparameter settings.
 - Requires regular monitoring of both generator and discriminator losses.
 - Takes much longer to train
- **VAE:**

- Generally exhibits more stable training because of the explicit regularization (KL divergence) and reconstruction loss.
- Takes less time to train
- Less resource intensive

c. Latent Space Representation

- **GAN:**
 - Does not explicitly enforce a structured latent space, which can make interpolation less predictable.
- **VAE:**
 - The latent space is smooth and interpretable, with clusters corresponding to different digit classes.

Potential Improvements via Hyperparameter Tuning

- **For GANs:**
 - Experiment with alternative loss functions and techniques like gradient penalty to reduce instability.
 - Fine-tuning learning rates between the generator and discriminator can further balance training.
- **For VAEs:**
 - Adjusting the weight of the KL divergence term can help improve image clarity.
 - Experimenting with the size of the latent space might yield a better trade-off between detail and smoothness.

Part 5: Anomaly Detection using the VAE

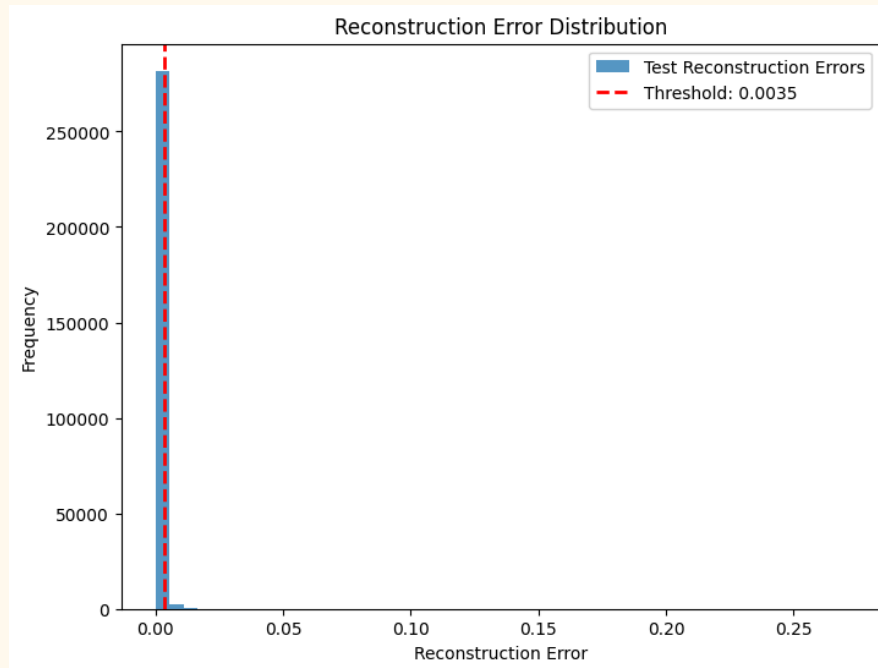
Approach

- The VAE is first trained on normal (non-anomalous) MNIST data.
- At inference time, the VAE reconstructs images, and the reconstruction error is used to flag anomalies.
 - Images with unusually high reconstruction errors are likely to be anomalies.

Insights

- **Effectiveness:**

- The VAE is effective at capturing the underlying data distribution. When faced with anomalous data, the reconstruction error increases noticeably.
- **Practical Application:**
 - This approach can be extended to real-world tasks such as fraud detection, system monitoring, or quality control.
- *Anomaly Detection Output:*



Discussion and Conclusion

Summary of Findings

- **Architectural Trade-offs:**
 - The GAN achieves high image fidelity with crisp details but is more challenging to train due to adversarial dynamics.
 - The VAE offers stable training and an interpretable latent space but may sacrifice some image sharpness.
- **Application Suitability:**
 - GANs are ideal when the primary goal is generating visually appealing images.
 - VAEs are better suited for tasks requiring latent space manipulation, such as image interpolation and anomaly detection.
- **Anomaly Detection:**

- The VAE's reconstruction error serves as a reliable metric for identifying anomalies, demonstrating its versatility beyond image generation.

Future Directions

- **Hybrid Models:**
 - Exploring hybrid approaches like VAE-GANs might combine the best of both worlds—high-quality images and structured latent spaces.
- **Extended Experiments:**
 - Further hyperparameter optimization and testing on varied datasets can yield additional insights.
- **Broader Applications:**
 - The methodologies demonstrated can be applied to other domains, including video, audio, and text generation, as well as to more complex anomaly detection scenarios.